



The evolution of microstructures and high temperature properties of $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ high entropy alloys



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ABSTRACT

The effects of Al and Ti contents on the microstructures and high temperature properties of $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ (x, y values in molar ratio, $x + y = 0.5$) high entropy alloys have been investigated. With an increase in Al content and a decrease in Ti concentrations, precipitated phases in the FCC high entropy γ matrix can evolve from η to γ' to β phases. Some of these alloys can possess high hardness values at elevated temperatures due to γ' precipitations, and continuous Cr_2O_3 can form on the surfaces of these alloys for protection against oxidation. This study can serve as a guideline for future high entropy alloy design for high temperature applications.

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1. Introduction

Without a based element, high entropy alloys can possess a variety of crystal structures, microstructures, and properties. Examples include FCC high entropy alloys, such as CoCrFeMnNi [1–3] and AlCoCrCuFeNi [4–6]; BCC high entropy alloys, such as WNbMoTaV [7,8] and HfNbTaTiZr [9–11]. High entropy alloys can exhibit high wear resistance [12,13], high hardness [13,14], significant high temperature compressive strength [8,10], and good corrosion resistance [15,16]. High entropy alloys can be single phase alloys [1–3,7,17], or they can contain several types of intermetallic phases, such as β [18,19], Laves [20,21] and σ [19,22]. Furthermore, high entropy alloys can exhibit sluggish diffusion effect to enhance the thermal stability of the microstructure [1,23,24].

Phases formed in multi-components alloy systems can be detrimental or beneficial to properties. For conventional superalloys, a type of high temperature phase instability associated with topologically closed packed (TCP) phases formation can cause deterioration on mechanical properties [25,26], this is due to the softening of the γ matrix resulting from depletion in strengthening elements such as W, Cr, Mo or Re [25,27]. On the other hand, the common strengthening precipitate in Fe–Ni and Ni-based superalloys is L_{12} γ' [28,29]. γ' precipitates are coherently embedded in γ

matrix, and it can hinder the motion of dislocations effectively [30]. Moreover, γ' particles at grain boundary can hinder grain boundary migration during thermal exposure [31]. In our previous work [32], $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ high entropy alloy can also contain γ' precipitates, but the γ' phase in this alloy is metastable and can transform into η phase. η phase is known to be brittle with little strain tolerance [33], so the transformation of γ' to η phase should be avoided. This work aims to optimize $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ high entropy alloy by studying the effect of Ti substitution by Al in $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ (x, y values in molar ratio, $x + y = 0.5$); this is to increase the thermal stability of γ' phase by substituting the η forming Ti by Al [32,33]. High temperature properties in terms of hardness and oxidation behaviors are presented and discussed. The present research can help the future alloy design of high entropy alloys toward high temperature structural applications.

2. Experimental methods

The compositions of interest in this work have included $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ ($\text{Ti}_{0.5}$), $\text{Al}_{0.1}\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.4}$ ($\text{Al}_{0.1}\text{Ti}_{0.4}$), $\text{Al}_{0.2}\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.3}$ ($\text{Al}_{0.2}\text{Ti}_{0.3}$), and $\text{Al}_{0.3}\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.2}$ ($\text{Al}_{0.3}\text{Ti}_{0.2}$), Table 1.

Samples of all alloys were prepared by arc-melting process in an argon atmosphere. As-cast samples were solution heat treated at 1150 °C for 6 h followed by water quench, then aged at various temperatures from 650 °C to 900 °C to precipitate γ' particles. All specimens were etched by 70% H_3PO_4 , and microstructures were

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Table 1
Composition of high entropy alloys in present study.

at%	Al	Co	Cr	Fe	Ni	Ti	γ'/η Solvus ($^{\circ}\text{C}$)	Solidus ($^{\circ}\text{C}$)	ΔS_{mix}
Ti ₀₅	—	27.3	18.2	18.2	27.3	9.1	1093	1212	−1.547R
Al ₀₁ Ti ₀₄	1.8	27.3	18.2	18.2	27.3	7.3	1053	1242	−1.592R
Al ₀₂ Ti ₀₃	3.6	27.3	18.2	18.2	27.3	5.5	1010	1289	−1.608R
Al ₀₃ Ti ₀₂	5.5	27.3	18.2	18.2	27.3	3.6	970	1323	−1.608R
Al ₀₅	9.1	27.3	18.2	18.2	27.3	—	—	1392	−1.547R

analyzed by Hitachi SU8010 scanning electron microscope (SEM) with an energy dispersive spectrometer (EDS). The volume fraction and size of γ' particles were estimated by ImageJ software. Heat treated samples were examined by transmission electron microscope (Tecnai F30 TEM) under 300 kV; specimens were prepared by twin-jet polishing in a solution of 10% HClO₄ in CH₃OH. X-ray diffractometer (SHIMADZU XRD-6000) equipped with Cu ($K\alpha$ radiation = 1.5406 Å) target was utilized to analyze phase constituents on samples with plate geometry.

Pyris Diamond TG/DTA was used to examine the thermal properties of samples of 30 mg in an Al₂O₃ crucible at temperatures from 25 to 1420 $^{\circ}\text{C}$ with a heating rate of 10 $^{\circ}\text{C}/\text{min}$ and 200 sccm of Ar flow. To measure the compositions of γ and γ' phases, microstructures had been coarsened by strain aging method, which was accomplished by firstly cold work on as-homogenized samples, then aged at 750 $^{\circ}\text{C}$ for 200 h to allow the coarsening of γ and γ' phases for EDS analysis.

Although alloys in present study are multi-phases HEA, precipitates can all be solutioned into single phase FCC matrix above the solvus temperatures of intermetallics. By referring to Boltzmann's hypothesis, the mixing entropy, ΔS_{mix} , of a n-element random solid-solution state can be calculated [34], and the ΔS_{mix} for each alloy is presented in Table 1:

$$\Delta S_{\text{mix}} = -R(X_A \ln X_A + X_B \ln X_B + \dots + X_n \ln X_n)$$

where X_A is the molar ratio of A element, X_B is the molar ratio of B element, etc. R is the gas constant (8.314 J/K mole).

Hardness of samples were measured by a high temperature Vickers hardness tester (Model AVK-HF, Akashi) equipped with a vacuum chamber and a heating unit. All specimens were tested at room temperature and from 500 $^{\circ}\text{C}$ to 1000 $^{\circ}\text{C}$ at an interval of 100 $^{\circ}\text{C}$ under the argon pressure close to 1 atm. A load of 49 N and a dwell time of 15 s was used. The heating rate of the sample was 10 $^{\circ}\text{C}/\text{min}$ and three indentations were made to get an average value. The indenter material was diamond, and indentation positions were all within individual grains. Isothermal oxidation tests were conducted at 900 $^{\circ}\text{C}$ for 200 h. All specimen surfaces were grinded by #1200 sand paper before oxidation test. The oxidation weight gains of specimens were measured by digital weighing scale, and cross sections were examined by SEM and EDS analysis. A sample of Inconel 718 sample, which was processed by the standard heat treatment (as solutioned + 720 $^{\circ}\text{C}/8$ h (furnace cooling) + 620 $^{\circ}\text{C}/8$ h (air cooling)) was used for comparative study.

3. Results and discussion

3.1. Microstructure evolution

Equilibrium phase constituents in these high entropy alloys can be described by Thermo-Calc (TCNI5 database) [35], Fig. 1(a). With an increase in Al content, the microstructures of these alloys can evolve from $\eta + \gamma$ to $\eta + \gamma + \gamma'$ to $\gamma + \gamma'$ to $\beta + \gamma + \gamma'$ to $\beta + \gamma$ for Ti₀₅, Al₀₁Ti₀₄, Al₀₂Ti₀₃, Al₀₃Ti₀₂, and Al₀₅, respectively; only Al₀₂Ti₀₃

can possess equilibrium γ and γ' two phases structure. XRD analysis on aged specimens (900 $^{\circ}\text{C}/50$ h) has confirmed the presence of η and β in the high Ti and high Al containing alloys, respectively, Fig. 1(b).

However, microstructure observations on samples aged at 750 $^{\circ}\text{C}$ for 50 h have revealed that main phase constituents in Ti₀₅ are γ and γ' , Fig. 2(a), which indicates sluggish γ' to η phase transformation in Ti₀₅. After Ti₀₅ has been aged at 900 $^{\circ}\text{C}$ for 50 h, plate-like η can be observed, Fig. 2(b). With an increase in Al content, phase stability of γ' can be enhanced, the as heat treated Al₀₁Ti₀₄ and Al₀₂Ti₀₃ can exhibit stable γ and γ' from 750 $^{\circ}\text{C}$ up to 900 $^{\circ}\text{C}$, Fig. 2(c) to (f). Although η has been presented as an equilibrium phase in Al₀₁Ti₀₄ according to Fig. 1, sluggish diffusion can

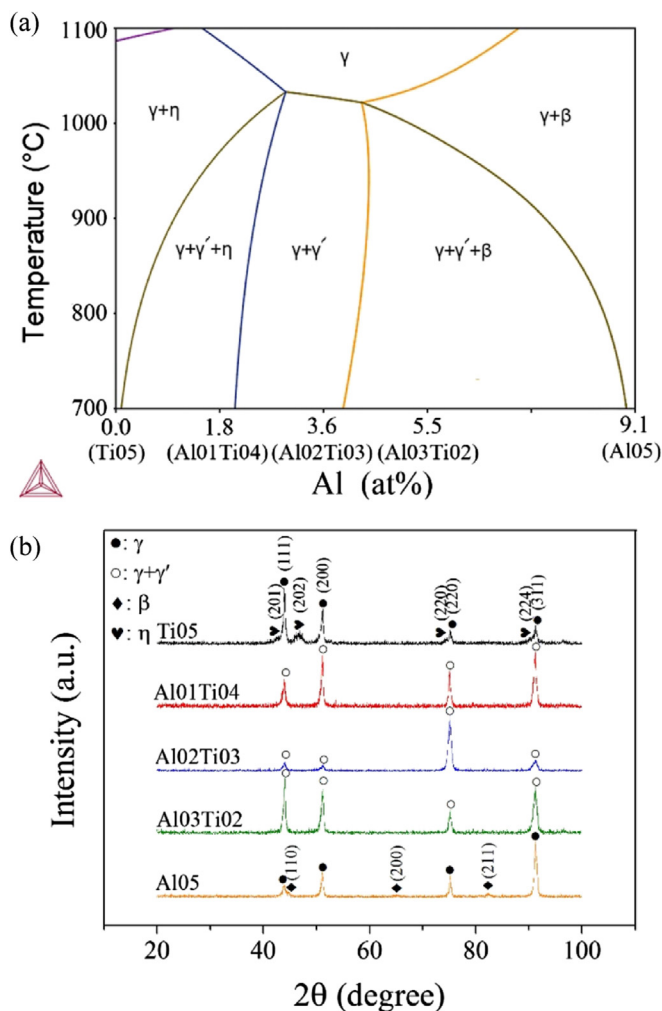


Fig. 1. (a) Phase diagram of Al_xCo₁₅CrFeNi₁₅Ti_{9-x} with varying Al content. (b) XRD analysis on high entropy alloys after 900 $^{\circ}\text{C}/50$ h aging.

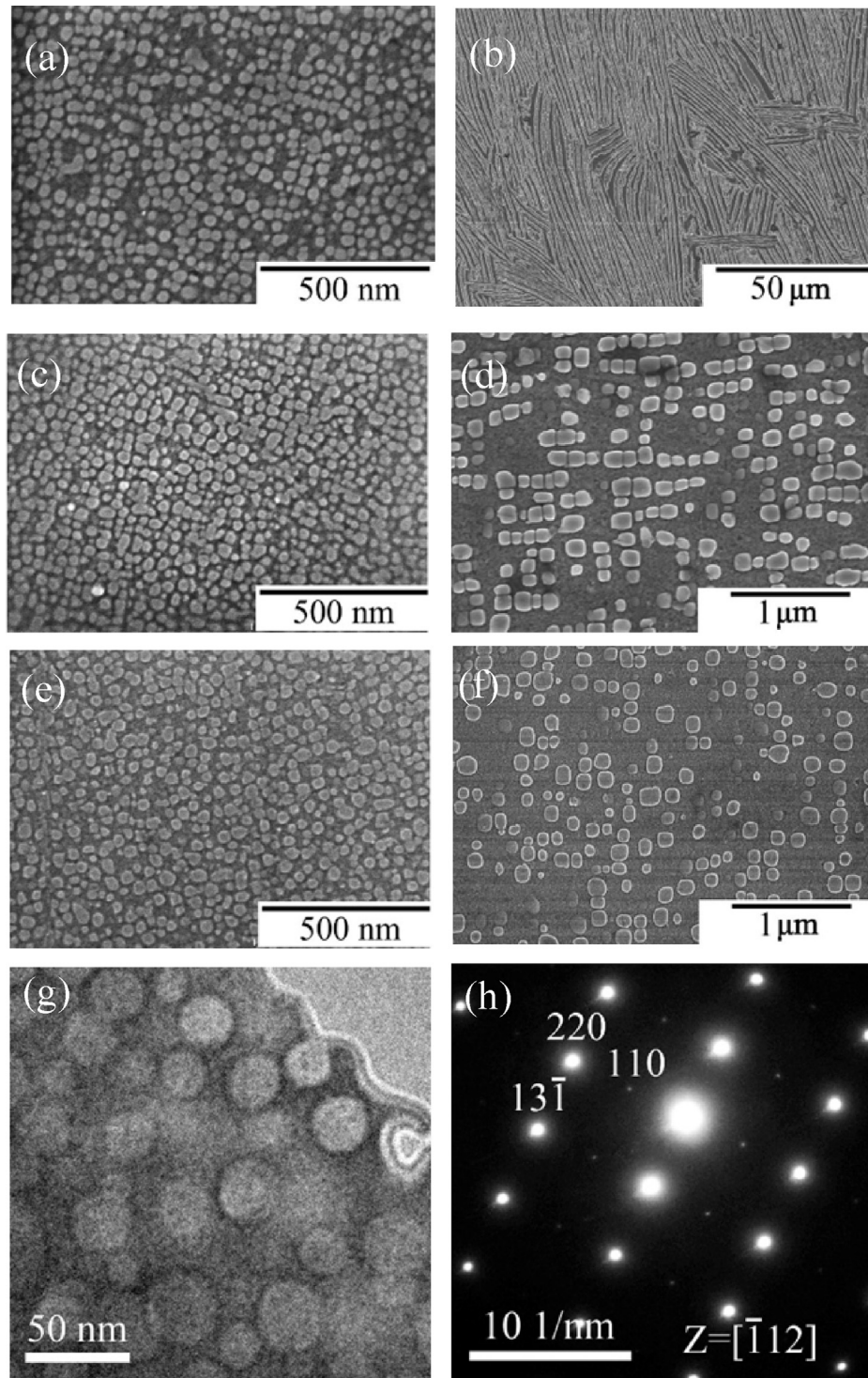


Fig. 2. Microstructures after aging for 50 h (a) Ti_{05} at 750 °C, (b) Ti_{05} at 900 °C, (c) $\text{Al}_{01}\text{Ti}_{04}$ at 750 °C, (d) $\text{Al}_{01}\text{Ti}_{04}$ at 900 °C, (e) $\text{Al}_{02}\text{Ti}_{03}$ at 750 °C, (f) $\text{Al}_{02}\text{Ti}_{03}$ at 900 °C, (g) $\text{Al}_{02}\text{Ti}_{03}$ at 750 °C, TEM bright field image, (h) $\text{Al}_{02}\text{Ti}_{03}$ at 750 °C, diffraction patterns.

render a microstructure containing no η phase, Fig. 2(d). Fig. 2(g) and (h) show a TEM micrograph of nano size γ' particles and its diffraction patterns in $\text{Al}_{02}\text{Ti}_{03}$ (aged at 750 °C for 50 h). The superlattice reflections of γ' can be observed, and the average size of γ' is 31 nm; the lattice misfit between γ and γ' has been estimated to be about 0.2%, which is confirmed by the spherical morphology of γ' particles shown in Fig. 2(g). Fig. 3(a) shows that

after ageing at 1000 °C for 50 h, $\text{Al}_{01}\text{Ti}_{04}$ contains needle-like η phase, and some γ' precipitates can still be observed; β phase in Al_{05} can be detected after 900 °C ageing for 50 h, Fig. 3(b). According to the compositions of γ and γ' shown in Table 2, the entropy term in γ' phase is increased with increasing Al substitution to Ti, this can destabilize γ' by decreasing the solvus temperature of γ' , Table 1. The increase in Al content has prevented the η phase

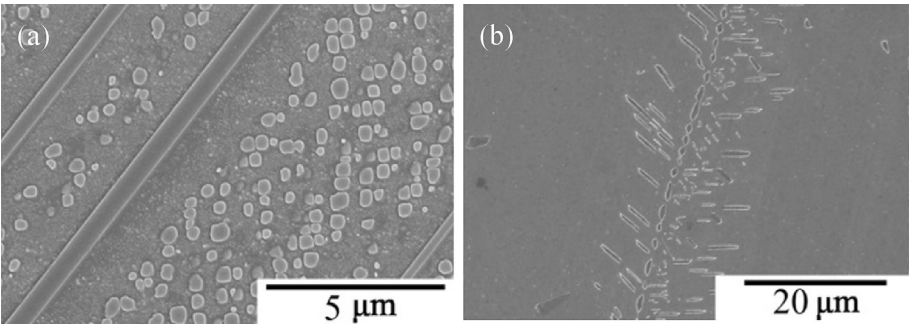


Fig. 3. Microstructures after aging for 50 h (a) Al₀₁Ti₀₄ at 1000 °C (contains needle like η and particulate γ'), and (b) Al₀₅ at 900 °C (brighter β phase precipitates).

Table 2
Composition and mixing entropy of γ and γ' in all alloys after 750 °C aging.

at%	Ti ₀₅		Al ₀₁ Ti ₀₄		Al ₀₂ Ti ₀₃		Al ₀₃ Ti ₀₂	
	γ	γ'	γ	γ'	γ	γ'	γ	γ'
Al	—	—	0.8	1.3	2.6	6.9	3.6	10.7
Co	27.8	17.0	31.8	16.7	32.8	19.0	30.0	18.6
Cr	29.4	12.9	19.4	7.0	22.6	8.5	20.2	9.4
Fe	24.4	11.9	20.8	7.3	19.8	9.0	20.4	9.6
Ni	15.2	34.6	23.4	51.4	19.0	44.8	23.4	44.8
Ti	3.2	17.0	3.8	16.2	2.8	11.3	2.4	6.8
ΔS_{mix}	−1.46R	−1.49R	−1.51R	−1.37R	−1.53R	−1.53R	−1.56R	−1.54R

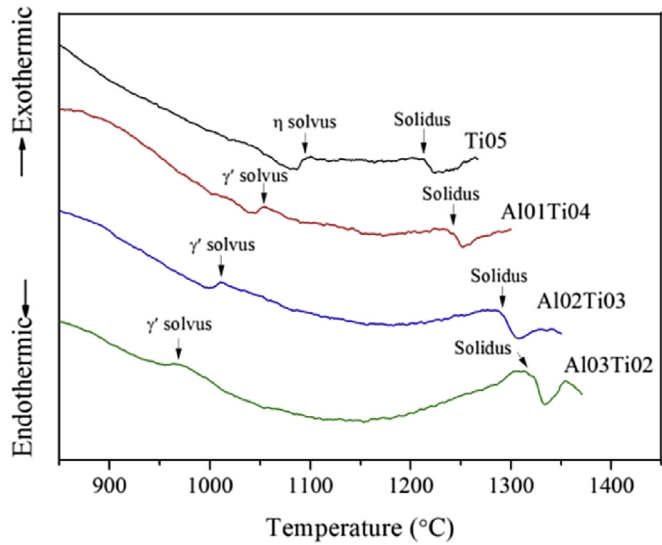


Fig. 4. Differential thermal analysis of Al_xCo_{1.5}CrFeNi_{1.5}Ti_y with varying Al content in heating runs.

formation, hence the thermal stability of γ' is improved in this respect. Comparing to conventional Ni-based superalloys, these high entropy alloys contain relatively high level of Fe content, however it can possess stable γ - γ' microstructure at high temperatures.

3.2. Thermal properties

Fig. 4 shows the results of DTA analysis on samples aged at 900 °C for 50 h, and the γ' solvus temperatures of heat treated samples are also listed in Table 1. The solvus temperatures of η and γ' phases, as well as the solidus temperatures have been verified with microstructure analysis on samples heat treated above and

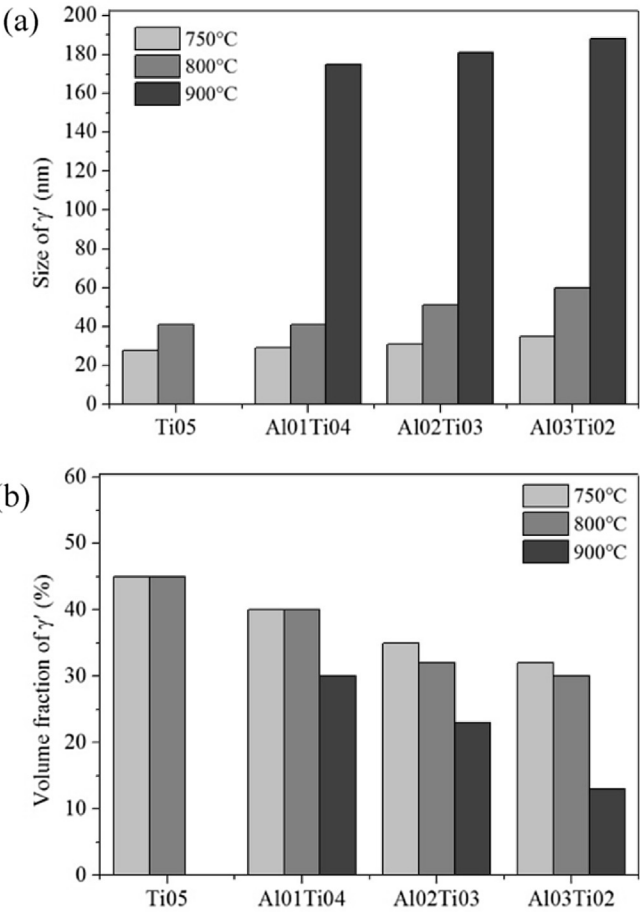


Fig. 5. (a) Mean size and (b) volume fraction of γ' of samples subjected to ageing at 750 °C, 800 °C, and 900 °C for 50 h.

below the temperatures given in Table 1. Al addition can eliminate η phase formation, and allows the equilibrium γ' solvus temperature to peak at 1010 °C, Fig. 1. However, higher Al/Ti ratio can decrease the solvus temperature of non-equilibrium γ' , increase the solidus, and induce the formation of detrimental β phase, Fig. 4. DTA has not been able to detect η solvus temperatures in $\text{Al}_{01}\text{Ti}_{04}$, which has been aged at 900 °C for 50 h, since diffusion in high entropy alloy is too sluggish to complete γ' to η phase transformation during DTA analysis. With an increase in Al content, the heat treatment window appears to be increased. The heat treatment window between equilibrium η solvus and solidus of $\text{Al}_{01}\text{Ti}_{04}$ is around 190 °C, which is actually similar to that of Inconel 718, i.e. 185 °C (the difference between δ solvus and solidus) [36]. The heat treatment window for $\text{Al}_{02}\text{Ti}_{03}$ is 279 °C. Although the substitution of Ti by Al can decrease the γ' solvus, the increase in solidus temperature can enlarge the heat treatment window, which will be beneficial for thermomechanical processing.

The size and volume fractions of γ' precipitates after various ageing conditions from 750 °C to 900 °C for 50 h are shown in Fig. 5, data of ageing temperatures below 750 °C are not listed since they are under-aged with γ' size smaller than 10 nm. It appears that an increase in Al/Ti ratio can increase the size of γ' and decrease its volume fraction. Furthermore, ageing at 800 °C and 900 °C can also

increase the size of γ' particles, and their volume fractions can be decreased as ageing temperatures are getting closer to γ' solvus temperatures. On the other hand, higher Ti content renders higher volume fraction of γ' and smaller size of γ' .

3.3. Hardness

Fig. 6(a) presents the room temperature hardness values of samples subjected to different ageing temperatures, and the results suggest that 750 °C is the optimum temperature for precipitation strengthening. All alloys have become harder after aging heat treatments, and the peak hardness can be achieved by the 750 °C ageing; alloys become softer when aging temperatures are above 750 °C. The softening phenomenon can be attributed by a decrease in the volume fraction of γ' as shown in Fig. 5(b). Furthermore, the sharp drop in strength of Ti_{05} can be a direct result of γ' to η phase transformation at 900 °C, which has been prevented by Al substitutions in other alloys, hence hardness values of $\text{Al}_{01}\text{Ti}_{04}$ and $\text{Al}_{02}\text{Ti}_{03}$ can be higher than that of Ti_{05} aged at 900 °C.

Hardness values from room temperature to 1000 °C are summarized in Fig. 6(b); all alloys have been aged at 750 °C for 50 h before hardness tests. Interestingly, Ti_{05} exhibits the highest hardness among all alloys. The unstable γ' in Ti_{05} can transform to brittle η at temperatures beyond 900 °C, however phase instability associated with η formation may not have much effect on the high temperature hardness tests due to short testing time. As Ti is being substituted by Al, the hardness of the alloy begins to drop. At room temperature, $\text{Al}_{01}\text{Ti}_{04}$ and Inconel 718 have similar hardness values; $\text{Al}_{01}\text{Ti}_{04}$ outperforms Inconel 718 at higher temperatures. $\text{Al}_{02}\text{Ti}_{03}$ and $\text{Al}_{03}\text{Ti}_{02}$ are softer than Inconel 718 up to 800 °C, interestingly, both of them outperform Inconel 718 at higher temperatures. The strengthening phase in Inconel 718 is γ'' , which can transform into detrimental δ phase at temperatures beyond 650 °C [37,38], so hardness of γ' bearing $\text{Al}_{02}\text{Ti}_{03}$ and $\text{Al}_{03}\text{Ti}_{02}$ can surpass that of Inconel 718 at higher temperatures. Although 1000 °C is around the γ' solvus temperatures of $\text{Al}_{02}\text{Ti}_{03}$ and $\text{Al}_{03}\text{Ti}_{02}$, the high temperature hardness testing times are again too short to fully dissolve all γ' into the matrix, hence γ' particles can still be present to provide strengthening. By referring to the physical metallurgy of γ' bearing superalloys [29,31,39,40], in addition to the core effects of high entropy alloys [41], the strength of these γ' bearing HEA can be affected by properties of γ and γ' , including phase stability, anti-phase boundary energy (APB), stacking fault energy, lattice misfit and volume fraction of γ' . High temperature hardness tests in present study reflect mainly the contributions from the intrinsic strength of γ and γ' , and volume fraction of γ' . The changes in γ' volume fractions agree with the high temperature hardness test results, Fig. 5(b) and Fig. 6(b). According to Table 2, The compositions of γ and γ' in high entropy alloys are highly alloyed comparing to those of conventional superalloys [42], furthermore, the high level of Ti content in γ' can increase the APB energy significantly to hinder the movement of dislocations [39,43], hence Ti_{05} and $\text{Al}_{01}\text{Ti}_{04}$ can exhibit higher resistance against high temperature indentation comparing to $\text{Al}_{02}\text{Ti}_{03}$.

For long term structural application at elevated temperatures, phase stability can be one of the most critical requirements. Future work will investigate creep resistances of these alloys; high entropy alloys can possess low stacking fault energy [44] and sluggish diffusion [1], both can be beneficial for high temperature creep resistance. Future work will also include the construction of time-temperature-transformation diagrams for phase transformation in these high entropy alloys.

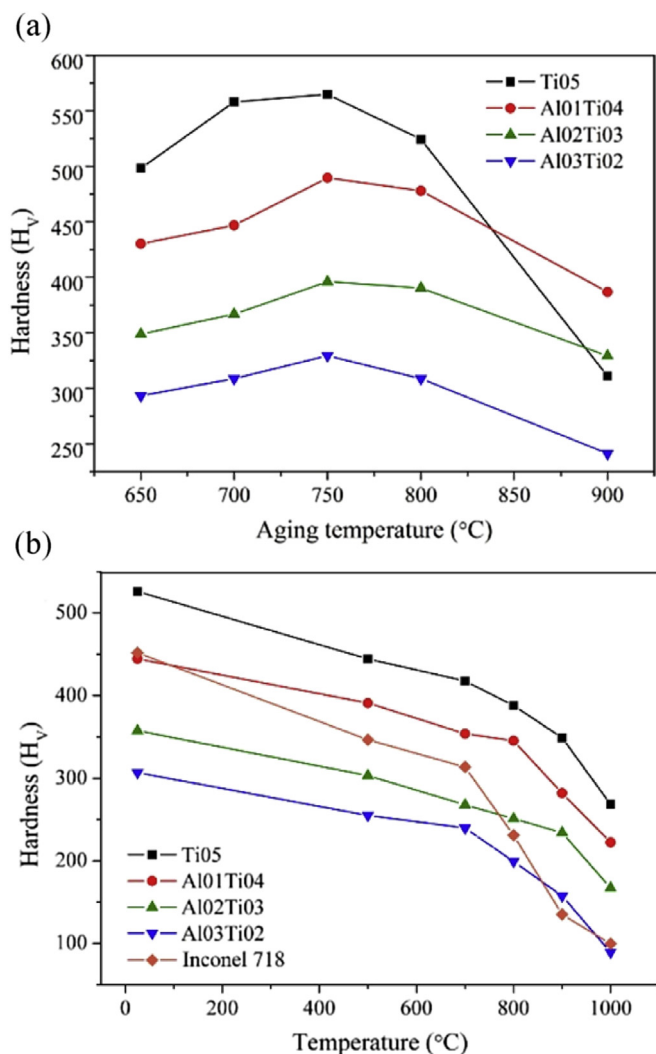


Fig. 6. (a) Room temperature hardness values of samples aged at various temperatures for 50 h, and (b) high temperature hardness values at different temperatures.

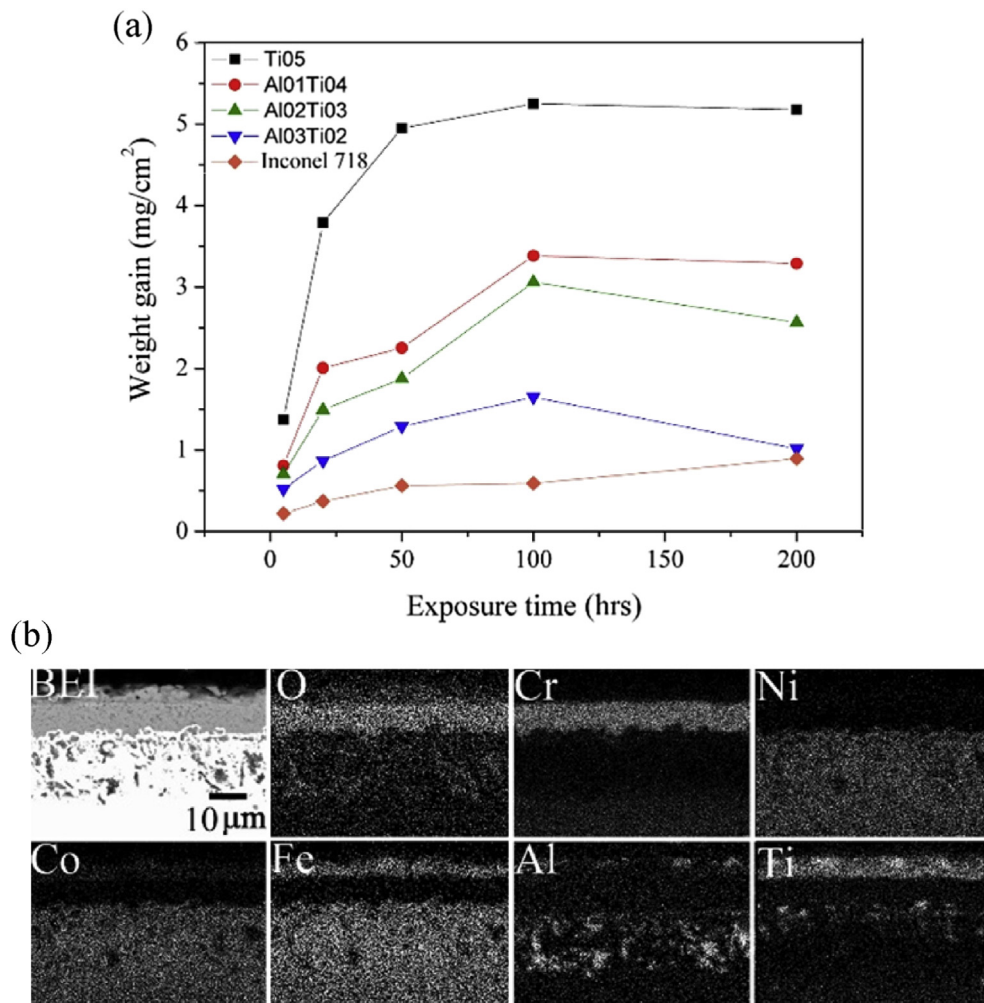


Fig. 7. (a) Oxidation weight gains at 900 °C, and (b) SEM image and EDS mappings of the cross section of Al_{0.2}Ti_{0.3} alloy after 900 °C/200 h oxidation.

3.4. High temperature oxidation properties

Fig. 7(a) presents the weight gain during isothermal oxidation tests at 900 °C for 200 h. Inconel 718 is a known chromia former [45], and has performed the best against oxidation comparing to other alloys in present study. The high entropy alloys can also exhibit certain degree of oxidation resistance with the formation of continuous chromia as shown in Fig. 7(b); EDS mapping indicates the formation of a dense and continuous Cr₂O₃ on the surface of Al_{0.2}Ti_{0.3}. The outermost oxides are iron oxide and titanium oxide mainly, and there are some internal oxidation of alumina beneath Cr₂O₃. The Al content can result in the “third element effect” [46] to increase the Cr activity and promote the formation of protective Cr₂O₃. However, sluggish diffusion may result delayed formation of continuous chromia layer, and high Ti content can also induce the formation of TiO₂ attributing to an increase in weight gain initially. Nevertheless, once continuous layer of Cr₂O₃ is formed, the oxidation can be hindered in high entropy alloys. In future alloy design, Cr and Al activity should be considered for more rapid formation of protective oxides.

4. Conclusions

The present study has investigated the effects of Al and Ti contents on the microstructures and high temperature properties

of Al_xCo_{1.5}CrFeNi_{1.5}Ti_y (x, y values in molar ratio, x + y = 0.5) high entropy alloys. Al substitution to part of Ti can eliminate detrimental η and enhance the thermal stability of γ'. These alloys can have good hardness from room temperature to high temperatures. Furthermore, continuous Cr₂O₃ can form to provide protection against high temperature oxidation. This work can serve as a guideline for future high entropy alloy design for high temperature applications.

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